CS5275 – The Algorithm Designer’s Toolkit (S2 AY2025/26)

Lecture 17:

Multiplicative weights update – applications

Learning a linear classifier

- Given m labeled examples: $(\mathbf{a}_1, \ell_1), (\mathbf{a}_2, \ell_2), \ldots, (\mathbf{a}_m, \ell_m)$
  - $\mathbf{a}_j \in \mathbb{R}^n$ is a feature vector
  - $\ell_j \in \{-1,1\}$ is a label.

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- Suppose there exists a vector $\mathbf{x} \in \mathbb{R}^n$ with $x_i \geq 0$ such that $\operatorname{sgn}(\mathbf{a}_j \cdot \mathbf{x}) = \ell_j$.
- **Goal:** Find a vector $\mathbf{x} \in \mathbb{R}^n$ with $x_i \geq 0$ such that $\operatorname{sgn}(\mathbf{a}_j \cdot \mathbf{x}) = \ell_j$.

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Without loss of generality, we may assume:
- $\mathbf{x} \cdot \mathbf{1} = 1$
- $\ell_j = 1$

$\mathbf{1}$ is the all-1 vector.

If this is not the case, then we may replace $\mathbf{a}_j$ with $-\mathbf{a}_j$.

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LP formulation:
- $\mathbf{a}_j \cdot \mathbf{x} > 0 \quad \forall j \in [m]$
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Claim: Suppose a large-margin solution $\mathbf{x}^*$ exists:

$$\mathbf{a}_j \cdot \mathbf{x}^* \geq \epsilon \quad \forall j \in [m]$$

Then there is an efficient algorithm that finds a feasible solution $\mathbf{x}$.

Connection to MWU

n coordinates

Multiplicative weights update:

- Assign initial weights to the experts to be identical weights.
- Update these weights multiplicatively and iteratively according to how well the experts performed.
- In each iteration, we decide the expert to follow based on the weighted distribution.

A candidate solution $\mathbf{x} \in \mathbb{R}^n$

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If all constraints $\mathbf{a}_j \cdot \mathbf{x} > 0$ are satisfied, then the problem is solved.

Otherwise, select an unsatisfied constraint and assign a cost to each coordinate based on how severely this constraint is violated.

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MWU performs competitively against any convex combination of experts.

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Algorithm

Multiplicative weights algorithm:

- Fix $0 < \eta \leq \frac{1}{2}$.
- Assign initial weights to the experts to be identical weights:
  - $w_i^{(1)} = 1$ for each expert $i \in \{1, \dots, n\}$.
- In each iteration $1 \leq t \leq T$:
  - Let $\Phi^{(t)} = \sum_{i=1}^{n} w_i^{(t)}$.
  - Select a distribution $\mathbf{p}^{(t)} = \left(\frac{w_1^{(t)}}{\Phi^{(t)}}, \frac{w_2^{(t)}}{\Phi^{(t)}}, \dots, \frac{w_n^{(t)}}{\Phi^{(t)}}\right)$.
  - Update the weight for every expert: $w_i^{(t+1)} = \left(1 + \frac{\eta m_i^{(t)}}{\rho}\right) w_i^{(t)}$.

This is seen as candidate solution $\mathbf{x} \in \mathbb{R}^n$.

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- If $\mathbf{x} = \mathbf{p}^{(t)}$ is already feasible, then we are done.
- Otherwise, there is an example $(\mathbf{a}_j, \ell_j)$ such that $\mathbf{a}_j \cdot \mathbf{p}^{(t)} \leq 0$.
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$\rho = \max_j \|\mathbf{a}_j\|_\infty$

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Run the multiplicative weights algorithm in the **gain** form with $m_i^{(t)} \in [-\rho, \rho]$.

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Suppose we are always in this case:

Analysis

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$$0 \geq \sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)}$$

Recall: $\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} \geq \sum_{t=1}^{T} \left(\mathbf{m}^{(t)} - \eta \left|\mathbf{m}^{(t)}\right|\right) \cdot \mathbf{p}^* - \frac{\rho \ln n}{\eta}$

the expected gain of the algorithm

$\mathbf{m}^{(t)} = \left(m_1^{(t)}, m_2^{(t)}, \ldots, m_n^{(t)}\right)$

- $m_i^{(t)} \in [-1,1]$ is the cost for expert i in iteration t.

$\mathbf{p}^{(t)}$ = distribution of experts in iteration t.

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$$\geq \sum_{t=1}^{T}\left(\mathbf{m}^{(t)} - \eta\left|\mathbf{m}^{(t)}\right|\right) \cdot \mathbf{x}^{*} - \frac{\rho \ln n}{\eta}$$

$$\geq T(\epsilon - \rho\eta) - \frac{\rho \ln n}{\eta}$$

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$m_i^{(t)} \in [-\rho, \rho] \rightarrow \left|\mathbf{m}^{(t)}\right| \cdot \mathbf{x}^{*} \leq \rho$

$\mathbf{a}_j \cdot \mathbf{x}^{*} \geq \epsilon \rightarrow \mathbf{m}^{(t)} \cdot \mathbf{x}^{*} \geq \epsilon$

Claim: Suppose a large-margin solution $\mathbf{x}^{*}$ exists:

$$\mathbf{a}_j \cdot \mathbf{x}^{*} \geq \epsilon \quad \forall j \in [m]$$

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Impossible

Suppose we are always in this case:

Analysis

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$$\geq T(\epsilon - \rho\eta) - \frac{\rho \ln n}{\eta} > 0$$

$\mathbf{m}^{(t)} = \left(m_1^{(t)}, m_2^{(t)}, \ldots, m_n^{(t)}\right)$

- $m_i^{(t)} \in [-1,1]$ is the cost for expert i in iteration t.

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Selecting $\eta = \frac{\epsilon}{2\rho}$ and $T = 1 + \left\lceil \frac{4\rho^2 \ln n}{\epsilon^2} \right\rceil$

Run MWU with these parameters. ✅

Claim: Suppose a large-margin solution $\mathbf{x}^*$ exists:

$$\mathbf{a}_j \cdot \mathbf{x}^* \geq \epsilon \quad \forall j \in [m]$$

Then there is an efficient algorithm that finds a feasible solution $\mathbf{x}$.

Summary of the main ideas

Experts	Coordinates
Decision = weighted distribution over the experts	Candidate solution
Outcome of an iteration	An unsatisfied constraint
Expected gain of the decision	How well the constraint is satisfied for the candidate solution
Gain of an expert	How well the constraint is satisfied on this coordinate

Zero-sum games

- Let $\mathbf{A}$ be the payoff matrix of a finite 2-player zero-sum game.

$\mathbf{A}$ is an $(n \times n)$ matrix.

$A(i,j) \in [0,1]$ specifies the payoff when the row player plays strategy $i \in [n]$ and the column player plays strategy $j \in [n]$.

If we do not have $A(i,j) \in [0,1]$ for all i and j, we may rescale them.

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	Rock	Paper	Scissors
Rock	0	1	-1
Paper	-1	0	1
Scissors	1	-1	0

Example: rock-paper-scissors

Zero-sum games

Goals:
- Column player wants to maximize the payoff.
- Row player wants to minimize the payoff.

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$A(i,j) \in [0,1]$ specifies the payoff when the row player plays strategy $i \in [n]$ and the column player plays strategy $j \in [n]$.

Suppose the row player chooses the mixed strategy

A probability distribution over the strategies

$$\mathbf{p} = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right).$$

A single strategy

The best response pure strategy of the column player is

$$j = \text{rock}.$$

The expected payoff is

$$A(\mathbf{p}, j) = \mathbf{E}_{i \sim \mathbf{p}}[A(i,j)] = \frac{1}{4} \cdot 0 + \frac{1}{4} \cdot (-1) + \frac{1}{2} \cdot 1 = \frac{1}{4}.$$

	Rock	Paper	Scissors
Rock	0	1	-1
Paper	-1	0	1
Scissors	1	-1	0

Example: rock-paper-scissors

Example 2

Goals:

- Column player wants to maximize the payoff.
- Row player wants to minimize the payoff.

$\mathbf{A}$ is an $(n \times n)$ matrix.

- Let $\mathbf{A}$ be the payoff matrix of a finite 2-player zero-sum game.

$A(i,j) \in [0,1]$ specifies the payoff when the row player plays strategy $i \in [n]$ and the column player plays strategy $j \in [n]$.

The best mixed strategy for the row player is

$$\mathbf{p} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right).$$

A probability distribution over the strategies

A single strategy

Regardless of the column pure strategy j, we have

$$A(\mathbf{p}, j) = \mathbf{E}_{i\sim\mathbf{p}}[A(i,j)] = 0.$$

	Rock	Paper	Scissors
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Example: rock-paper-scissors

Max–min inequality

Goals:

- Column player wants to maximize the payoff.
- Row player wants to minimize the payoff.

$$\min_{\mathbf{p}} \max_{\mathbf{q}} A(\mathbf{p}, \mathbf{q}) = \min_{\mathbf{p}} \max_{j} A(\mathbf{p}, j) \geq \max_{\mathbf{q}} \min_{i} A(i, \mathbf{q}) = \max_{\mathbf{q}} \min_{\mathbf{p}} A(\mathbf{p}, \mathbf{q})$$

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Responding with a mixed strategy $\mathbf{q}$.

Responding with a pure strategy j.

Similar

Intuition: For the player that decides its strategy later, **randomness** does not help.

The row player first announces its mixed strategy $\mathbf{p}$.
The column player decides the best response to $\mathbf{p}$.

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$$\max_{j} A(\mathbf{p},j) \geq \max_{\mathbf{q}} A(\mathbf{p},\mathbf{q})$$

$$\max_{j} A(\mathbf{p},j) \leq \max_{\mathbf{q}} A(\mathbf{p},\mathbf{q})$$

Each pure strategy j is also a mixed strategy $\mathbf{q}$.

For each mixed strategy $\mathbf{q}$, there exists a pure strategy j such that $A(\mathbf{p},j) \geq A(\mathbf{p},\mathbf{q})$.

$A(\mathbf{p},\mathbf{q})$ is a convex combination of $A(\mathbf{p},j)$ over all strategies j.

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Intuition: The player that decides its strategy **later** has advantage.

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- Row player wants to minimize the payoff.

$$\min_{\mathbf{p}} \max_{\mathbf{q}} A(\mathbf{p},\mathbf{q}) = \underbrace{\min_{\mathbf{p}} \max_{j} A(\mathbf{p},j)} \geq \underbrace{\max_{\mathbf{q}} \min_{i} A(i,\mathbf{q})} = \max_{\mathbf{q}} \min_{\mathbf{p}} A(\mathbf{p},\mathbf{q})$$

The row player first announces its mixed strategy $\mathbf{p}$.
The column player decides the best response j to $\mathbf{p}$.

The column player first announces its mixed strategy $\mathbf{q}$.
The row player decides the best response i to $\mathbf{q}$.

Intuition: The player that decides its strategy **later** has advantage.

Max–min inequality

Goals:

- Column player wants to maximize the payoff.
- Row player wants to minimize the payoff.

$$\min_{\mathbf{p}} \max_{\mathbf{q}} A(\mathbf{p},\mathbf{q}) = \min_{\mathbf{p}} \max_{j} A(\mathbf{p},j) \geq \max_{\mathbf{q}} \min_{i} A(i,\mathbf{q}) = \max_{\mathbf{q}} \min_{\mathbf{p}} A(\mathbf{p},\mathbf{q})$$

Select $\mathbf{p}^*$ and $\mathbf{q}^*$ to be the best row and column mixed strategies:

- $\max_{\mathbf{q}} A(\mathbf{p}^*,\mathbf{q}) = \min_{\mathbf{p}} \max_{\mathbf{q}} A(\mathbf{p},\mathbf{q})$
- $\min_{\mathbf{p}} A(\mathbf{p},\mathbf{q}^*) = \max_{\mathbf{q}} \min_{\mathbf{p}} A(\mathbf{p},\mathbf{q})$

$$\min_{\mathbf{p}} \max_{\mathbf{q}} A(\mathbf{p},\mathbf{q}) = \max_{\mathbf{q}} A(\mathbf{p}^*,\mathbf{q}) \geq A(\mathbf{p}^*,\mathbf{q}^*) \geq \min_{\mathbf{p}} A(\mathbf{p},\mathbf{q}^*) = \max_{\mathbf{q}} \min_{\mathbf{p}} A(\mathbf{p},\mathbf{q})$$

Minimax theorem

Goals:

- Column player wants to maximize the payoff.
- Row player wants to minimize the payoff.

$$\min_{\mathbf{p}} \max_{\mathbf{q}} A(\mathbf{p},\mathbf{q}) = \min_{\mathbf{p}} \max_{j} A(\mathbf{p},j) \geq \max_{\mathbf{q}} \min_{i} A(i,\mathbf{q}) = \max_{\mathbf{q}} \min_{\mathbf{p}} A(\mathbf{p},\mathbf{q})$$

Minimax theorem:

$$\underbrace{\lambda^* = \min_{\mathbf{p}} \max_{j} A(\mathbf{p},j)}_{\text{Ddefinition of } \lambda^*} = \max_{\mathbf{q}} \min_{i} A(i,\mathbf{q})$$

Minimax theorem

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$$\lambda^* = \underbrace{\min_{\mathbf{p}} \max_{j} A(\mathbf{p}, j)}_{\text{Ddefinition of } \lambda^*} = \max_{\mathbf{q}} \min_{i} A(i, \mathbf{q})$$

Claim: There is an efficient algorithm that estimates λ^* within an additive $\pm\epsilon$ error and finds near−optimal mixed strategies $\tilde{\mathbf{p}}$ and $\tilde{\mathbf{q}}$ of the two players.

- $\lambda^* - \epsilon \leq \min_{i} A(i, \tilde{\mathbf{q}})$
- $\max_{j} A(\tilde{\mathbf{p}}, j) \leq \lambda^* + \epsilon.$

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Minimax theorem:

$$\lambda^* = \underbrace{\min_{\mathbf{p}} \max_{j} A(\mathbf{p},j)}_{\text{Ddefinition of } \lambda^*} = \max_{\mathbf{q}} \min_{i} A(i,\mathbf{q})$$

$\epsilon \to 0$

$$\min_{\mathbf{p}} \max_{j} A(\mathbf{p},j) - \epsilon$$
$$= \lambda^* - \epsilon$$
$$\leq \min_{i} A(i,\tilde{\mathbf{q}})$$
$$\leq \max_{\mathbf{q}} \min_{i} A(i,\mathbf{q})$$

Claim: There is an efficient algorithm that estimates λ^* within an additive $\pm\epsilon$ error and finds near−optimal mixed strategies $\tilde{\mathbf{p}}$ and $\tilde{\mathbf{q}}$ of the two players.
- $\lambda^* - \epsilon \leq \min_{i} A(i,\tilde{\mathbf{q}})$
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Algorithm

Goals:

- Column player wants to maximize the payoff.
- Row player wants to minimize the payoff.

Multiplicative weights algorithm:

- Fix $0 < \eta \leq \frac{1}{2}$.
- Assign initial weights to the experts to be identical weights:
  - $w_i^{(1)} = 1$ for each expert $i \in \{1, \dots, n\}$.
- In each iteration $1 \leq t \leq T$:
  - Let $\Phi^{(t)} = \sum_{i=1}^{n} w_i^{(t)}$.
  - Select a distribution $\mathbf{p}^{(t)} = \left(\frac{w_1^{(t)}}{\Phi^{(t)}}, \frac{w_2^{(t)}}{\Phi^{(t)}}, \dots, \frac{w_n^{(t)}}{\Phi^{(t)}}\right)$.
  - Update the weight for every expert: $w_i^{(t+1)} = \left(1 - \eta m_i^{(t)}\right) w_i^{(t)}$.

Treat each row pure strategy $\{1,2, \dots, n\}$ as an expert.

This is seen as a row mixed strategy.

Algorithm

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- Pick $j^{(t)}$ as a best response column pure strategy against $\mathbf{p}^{(t)}$.
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Expected cost $= \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} = A\left(\mathbf{p}^{(t)}, j^{(t)}\right) = \max_j A(\mathbf{p}^{(t)}, j) \geq \min_{\mathbf{p}} \max_j A(\mathbf{p}, j) = \lambda^*$

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If optimal row mixed strategy $\mathbf{p}^*$ is used, then $\mathbf{m}^{(t)} \cdot \mathbf{p}^* = A\left(\mathbf{p}^*, j^{(t)}\right) \leq \max_j A(\mathbf{p}^*, j) = \min_{\mathbf{p}} \max_j A(\mathbf{p}, j) = \lambda^*$

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Algorithm

Goals:
- Column player wants to maximize the payoff.
- Row player wants to minimize the payoff.

Intuitively: MWU performs competitively against any convex combination of experts. → If we run the algorithm for many iterations, then on average, the payoff of a row mixed strategy $\mathbf{p}^{(t)}$ is very close to λ^*.

Expected cost $= \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} = A\left(\mathbf{p}^{(t)}, j^{(t)}\right) = \max_j A(\mathbf{p}^{(t)}, j) \geq \min_{\mathbf{p}} \max_j A(\mathbf{p}, j) = \lambda^*$

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Analysis

$$\lambda^* \le \frac{1}{T}\sum_{t=1}^{T} A\left(\mathbf{p}^{(t)}, j^{(t)}\right) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)}$$

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Analysis

$\mathbf{p}'$ is any row mixed strategy

$$\lambda^* \le \frac{1}{T}\sum_{t=1}^{T} A\left(\mathbf{p}^{(t)}, j^{(t)}\right) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} \le \frac{\ln n}{\eta T} + \eta + \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}'$$

Recall: $\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} \le \underbrace{\sum_{t=1}^{T}\left(\mathbf{m}^{(t)} + \eta\left|\mathbf{m}^{(t)}\right|\right) \cdot \mathbf{p}^*}_{\le \eta T + \sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}^*} + \frac{\rho \ln n}{\eta}$

$\rho = 1$

$\boxed{\text{Expected cost} = \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} = A\left(\mathbf{p}^{(t)}, j^{(t)}\right) = \max_j A\left(\mathbf{p}^{(t)}, j\right) \ge \min_{\mathbf{p}} \max_j A(\mathbf{p}, j) = \lambda^*}$

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Analysis

$$\lambda^* \le \frac{1}{T}\sum_{t=1}^{T} A\left(\mathbf{p}^{(t)}, j^{(t)}\right) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} \le \frac{\ln n}{\eta T} + \eta + \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}'$$

$$= \frac{\ln n}{\eta T} + \eta + \frac{1}{T}\sum_{t=1}^{T} A\left(\mathbf{p}', j^{(t)}\right)$$

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Analysis

$$\lambda^* \le \frac{1}{T}\sum_{t=1}^{T} A\left(\mathbf{p}^{(t)}, j^{(t)}\right) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} \le \frac{\ln n}{\eta T} + \eta + \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}'$$

$$= \frac{\ln n}{\eta T} + \eta + \frac{1}{T}\sum_{t=1}^{T} A\left(\mathbf{p}', j^{(t)}\right)$$

$$\eta = \frac{\epsilon}{2}$$

$$T = \lceil 4\epsilon^{-2} \ln n \rceil$$

$$\le \epsilon + \frac{1}{T}\sum_{t=1}^{T} A\left(\mathbf{p}', j^{(t)}\right)$$

Expected cost $= \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} = A\left(\mathbf{p}^{(t)}, j^{(t)}\right) = \max_j A(\mathbf{p}^{(t)}, j) \ge \min_{\mathbf{p}} \max_j A(\mathbf{p}, j) = \lambda^*$

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Analysis

$$\lambda^* \leq \frac{1}{T}\sum_{t=1}^{T} A\left(\mathbf{p}^{(t)}, j^{(t)}\right) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} \leq \frac{\ln n}{\eta T} + \eta + \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}'$$

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$\eta = \frac{\epsilon}{2}$

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$$\leq \epsilon + \frac{1}{T}\sum_{t=1}^{T} A\left(\mathbf{p}', j^{(t)}\right)$$

$$\leq \epsilon + \lambda^*$$

$\mathbf{p}' = \mathbf{p}^*$

If optimal row mixed strategy $\mathbf{p}^*$ is used, then $\mathbf{m}^{(t)} \cdot \mathbf{p}^* = A\left(\mathbf{p}^*, j^{(t)}\right) \leq \max_j A(\mathbf{p}^*, j) = \min_{\mathbf{p}} \max_j A(\mathbf{p}, j) = \lambda^*$

The **cost** vector $\mathbf{m}^{(t)} = \left(m_1^{(t)}, m_2^{(t)}, \ldots, m_n^{(t)}\right)$ is defined as follows:

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$\mathbf{p}'$ is any row mixed strategy

Analysis

$$\lambda^* \le \frac{1}{T}\sum_{t=1}^{T} A\left(\mathbf{p}^{(t)}, j^{(t)}\right) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} \le \frac{\ln n}{\eta T} + \eta + \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}'$$

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$\eta = \frac{\epsilon}{2}$

$T = \lceil 4\epsilon^{-2} \ln n \rceil$

$$\le \epsilon + \frac{1}{T}\sum_{t=1}^{T} A\left(\mathbf{p}', j^{(t)}\right)$$

$$\le \epsilon + \lambda^*$$

Near-optimal row mixed strategy: There exists $t \in [T]$ such that $\max_j A(\mathbf{p}^{(t)}, j) = A\left(\mathbf{p}^{(t)}, j^{(t)}\right) \le \lambda^* + \epsilon$.

Row player wants to minimize the payoff.

The **cost** vector $\mathbf{m}^{(t)} = \left(m_1^{(t)}, m_2^{(t)}, \dots, m_n^{(t)}\right)$ is defined as follows:

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$$\lambda^* \le \frac{1}{T}\sum_{t=1}^{T} A\left(\mathbf{p}^{(t)}, j^{(t)}\right) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} \le \frac{\ln n}{\eta T} + \eta + \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}'$$

$$= \frac{\ln n}{\eta T} + \eta + \frac{1}{T}\sum_{t=1}^{T} A\left(\mathbf{p}', j^{(t)}\right)$$

$\eta = \frac{\epsilon}{2}$

$T = \lceil 4\epsilon^{-2} \ln n \rceil$

$$\le \epsilon + \frac{1}{T}\sum_{t=1}^{T} A\left(\mathbf{p}', j^{(t)}\right)$$

$$= \epsilon + A(\mathbf{p}', \tilde{\mathbf{q}})$$

$\tilde{\mathbf{q}}$ is defined as the distribution that assigns j with probability $\frac{|\{t : j^{(t)} = j\}|}{T}$.

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$$\lambda^* \leq \frac{1}{T}\sum_{t=1}^{T} A\left(\mathbf{p}^{(t)}, j^{(t)}\right) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} \leq \frac{\ln n}{\eta T} + \eta + \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}'$$

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$\tilde{\mathbf{q}}$ is defined as the distribution that assigns j with probability $\frac{|\{t : j^{(t)} = j\}|}{T}$.

Near-optimal column mixed strategy: $\lambda^* - \epsilon \leq A(\mathbf{p}', \tilde{\mathbf{q}}) = \min_i A(i, \tilde{\mathbf{q}})$

The analysis works for any $\mathbf{p}'$.

References

- Section 3 of “The Multiplicative Weights Update Method: a Meta-Algorithm and Applications” by Sanjeev Arora, Elad Hazan, and Satyen Kale
  - https://theoryofcomputing.org/articles/v008a006/